XINYU LI

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Multimodal foundation-model researcher with 8+ years building VLMs, VLAs, and omni models. MTS at Physical Intelligence working on π-family models; previously Principal Scientist at Amazon AGI leading the core multimodal team of 15 building the Nova Family. H-index 28 with oral and spotlight papers at CVPR, ECCV, and NeurIPS.

EXPERIENCE

Physical Intelligence — *Member of Technical Staff* 2026 – Present *San Francisco, CA*

- **VLM Pretraining.** Developing next generation π-family VLM base models for high-level multimodal understanding, powering task planning, reward modeling, auto-annotation, and behavior/data analysis.
- **VLA Pretraining.** Develop π-family VLA base models for robotic manipulation, grounding scene understanding and object-centric reasoning into task-conditioned physical actions.

AGI, Amazon — *Principal Applied Scientist* 2022 – 2025 *Bellevue, WA*

- **Amazon Nova Multimodal Models.** Led a 15-person team building the Amazon Nova/Nova 2 multimodal stack, including visual encoders, multimodal embeddings, cross-modal reasoning, retrieval/RAG, and multimodal LLM capabilities for production deployment.
- **Training Infrastructure and Research.** Built end-to-end infrastructure for hundred-billion-parameter-scale multimodal training across pretraining, post-training, evaluation, and serving; led video/multimodal research with publications at CVPR, ICCV, ECCV, NeurIPS, and ICLR.

AML, ByteDance — *Staff Research Scientist* 2021 – 2022 *Bellevue, WA*

- **Multimodal Modeling.** Led vision–audio–language modeling for compliance, search, and recommendation; published at CVPR, ICLR, and ICASSP.

AWS AI, Amazon — *Senior Applied Scientist* 2018 – 2021 *Seattle, WA*

- **Rekognition Video.** Led research and production delivery for video classification, tracking, and segmentation, serving Prime Video, Amazon Stores, and AWS customers.
- **Research and Open Source.** Published CVPR Oral TubeR, ECCV Spotlight Directional Temporal Modeling, ICCV VidTr, and NeurIPS Spotlight LSTR; core contributor to GluonCV and GluonMM.

RESEARCH OUTPUT

Full list: Google Scholar | Homepage. * Equal contribution.

Foundation Model Reports

- **2026** · π0.7: A Steerable Model with Emergent Capabilities. Physical Intelligence.
- **2025** · Amazon Nova 2: Multimodal Reasoning and Generation Models. Amazon AGI.
- **2025** · Amazon Nova Multimodal Embeddings. Amazon AGI.
- **2024** · Amazon Nova Family / Nova Premier. Amazon AGI.

Selected Publications

- **CVPR26.** STORM: End-to-End Referring Multi-Object Tracking in Videos. Z. Lu, J. Yi, J. Wang, Y. Chen, J. Chen, **X. Li**, D. Modolo.
- **ICCV25.** SemiVisBooster: Boosting Semi-Supervised Learning for Fine-Grained Classification through Pseudo-Label Semantic Guidance. W. Zhang, **X. Li**, C. Gao, I. Marsic.
- **NeurIPS24.** Video Token Merging for Long Video Understanding. S.-H. Lee, J. Wang, Z. Zhang, D. Fan, **X. Li**.
- **ECCV24.** Text-Guided Video Masked Autoencoder. D. Fan, J. Wang, S. Liao, Z. Zhang, V. Bhat, **X. Li**.

- **ICCV23.** Motion-Guided Masking for Spatiotemporal Representation Learning. D. Fan, J. Wang, S. Liao, Y. Zhu, V. Bhat, H. Santos-Villalobos, R. MV, **X. Li**.
- **CVPR23.** Revisiting Multimodal Representation in Contrastive Learning: From Patch and Token Embeddings to Finite Discrete Tokens. Y. Chen, J. Yuan, Y. Tian, S. Geng, **X. Li**, D. Zhou, D.N. Metaxas, H. Yang.
- **CVPR22 Oral.** TubeR: Tubelet Transformer for Video Action Detection. J. Zhao*, Y. Zhang*, **X. Li***, H. Chen, B. Shuai, M. Xu, C. Liu, K. Kundu, Y. Xiong, D. Modolo, I. Marsic, C.G.M. Snoek, J. Tighe.
- **CVPR22 Oral.** What to Look at and Where: Semantic and Spatial Refined Transformer for Detecting Human–Object Interactions. A.S.M. Iftekhar, H. Chen, K. Kundu, **X. Li**, J. Tighe, D. Modolo.
- **CVPR22 Oral.** Stochastic Backpropagation: A Memory Efficient Strategy for Training Video Models. F. Cheng, M. Xu, Y. Xiong, H. Chen, **X. Li**, W. Li, W. Xia.
- **ICCV21.** VidTr: Video Transformer Without Convolutions. Y. Zhang*, **X. Li***, C. Liu, B. Shuai, Y. Zhu, B. Brattoli, H. Chen, I. Marsic, J. Tighe.
- **NeurIPS21 Spotlight.** Long Short-Term Transformer for Online Action Detection. M. Xu, Y. Xiong, H. Chen, **X. Li**, W. Xia, Z. Tu, S. Soatto.
- **ECCV20 Spotlight.** Directional Temporal Modeling for Action Recognition. **X. Li**, B. Shuai, J. Tighe.

Patents

- **US12526485B1 (2026).** Content aware graphical subtitles. H. Mahyar, J.C. Willeford, A. Cholkar, **X. Li**.
- **US12387097B2 (2025).** Efficient video processing via temporal progressive learning. P. Wang, H. Wang, Xianhang Li, **Xinyu Li**.
- **US11423265B1 (2022).** Content moderation using object detection and image classification. H. Chen, H. Wu, H. Li, M.Q.T. Lam, **X. Li**, K. Kundu, M. Wang, J.P. Tighe, R. Bhotika.

Open Source

- **GluonCV/GluonMM.** Core contributor to Gluon toolkits; GluonCV has 5.9K+ GitHub stars.
- **TubeR.** Paper | Code. Official implementation of the Tubelet Transformer for video action detection.
- **STORM-Bench.** Paper | Code. Benchmark and code for MLLM-based referring multi-object tracking.

INVITED TALKS

- **2026.** *A Causal History of Video Understanding.* University of Tennessee, Knoxville, Department of Electrical Engineering and Computer Science. Invited by Dr. Hector Santos-Villalobos.
- **2025.** *Referral Multi-object Tracking.* Scale AI research. Invited by Dr. Yuzhong He.
- **2023.** *Large Multimodal Models and Their Applications.* Amazon Prime Video Tech Week.
- **2021.** *Transformers and Their Applications in Video Understanding.* DAMO Academy, Alibaba Group (U.S.). Invited by Dr. Pichao Wang.

SKILLS

- **Foundation Models.** VLM, VLA, MLLM, and omni-model pretraining/post-training; data mixture design, model scaling, SFT/RL, reward modeling, auto-annotation, and evaluation.
- **Multimodal and Embodied AI.** Multimodal embeddings, retrieval/RAG, long-context video, scene/object grounding, tracking, segmentation, action grounding, and robot manipulation.
- **Scale and Leadership.** Hundred-billion-parameter-scale multimodal training, PyTorch/Megatron-LM, 3D parallelism, production training/serving pipelines, science team leadership, and mentoring.

PROFESSIONAL SERVICE

Program Committee. CVPR, ICCV, ECCV, NeurIPS, ICLR, ACM MM, INTERSPEECH, ICASSP.
Journal Reviewer. TPAMI, IJCV, TIP.

EDUCATION

Ph.D., Computer Engineering	Rutgers University	2013 – 2018
B.Eng., Communication Engineering	UESTC	2009 – 2013